Advanced Trade Studies and Decision Analysis Methods

Game Theory

Game Theory

- Life is full of conflict and competition.
  - Adversaries in conflict include parlor games, military battles, political campaigns, advertising and marketing campaigns by competing business firms, etc.
  - Final outcome depends primarily upon the combination of strategies selected by the adversaries.
- Game theory is a mathematical theory that deals with the general features of competitive situations in a formal, abstract way.
  - Applications in various areas, including decision making.

Game Theory

- Simple case: two-person, zero-sum games.
  - Involves only two adversaries or players.
  - Called zero-sum games because one player wins whatever the other one loses, so that the sum of their net winnings is zero.

Two-Person, Zero-Sum Games

- Odds and evens.
  - Two players.
  - Each player simultaneously selects number 1 or 2.
  - The numbers from both players are added together.
  - Consider player 1 wins when the outcome is even.
  - Then, player 2 wins when the outcome is odd.
  - The bet is $1.
  - If the total number is even, then player 1 wins.
  - If the total number is odd, then player 2 wins.

Two-Person, Zero-Sum Games

- Odds and evens.
  - Each player has two strategies: choose number 1 or 2.
  - Payoff table:

Strategy		**Player 2**	
		1	2
Player 1	1	1	-1
	2	-1	1

Strategy		**Player 1**	
		1	2
Player 2	1	-1	1
	2	1	-1

Two-Person, Zero-Sum Games

- Characterized by:
  - Strategies of player 1.
  - Strategies of player 2.
  - Payoff table.

- Before the game begins, each player knows the available strategies, the ones the opponent has available, and the payoff table.
- Actual play of the game consists of each player simultaneously choosing a strategy without knowing the opponent’s choice.

Two-Person, Zero-Sum Games

- Strategies:
  - One simple action.
  - Predetermined rule that specifies completely how one intends to respond to each possible circumstance at each stage of the game.
- Payoff table:
  - Shows the gain (positive or negative) that would result from each combination of strategies for the two players.

Two-Person, Zero-Sum Games

- Rational criteria for selecting a strategy.
  - Both players are rational.
  - Both players choose their strategies solely to promote their own welfare (no compassion for the component).

Example – Problem Formulation

- Two players and three identical strategies for each player.
- Both players use the same formulation for choosing their strategies.
- Payoff table: value of choosing one of the strategy for a player when the other player chooses their own strategies.

Example – Payoff Table

- From Player 1 viewpoint.

Strategy		Player 2		
		1	2	3
Player 1	1			
	2			
	3			

- Use 3 alternative sets of data for the payoff table to illustrate how to solve 3 different types of games.

Example – Dominated Strategies

- A strategy is dominated by a second strategy if the second strategy is always at least as good (and sometimes better) regardless of what the opponent does.
  - A dominated strategy can be eliminated immediately from further consideration.

Strategy		Player 2		
		1	2	3
Player 1	1	1	2	4
	2	1	0	5
	3	0	1	-1

Example – Dominated Strategies

- From Player 1 viewpoint:
  - Strategy 3 is dominated by Strategy 1.

Strategy		Player 2		
		1	2	3
Player 1	1	1	2	4
	2	1	0	5

- From Player 2 viewpoint:
  - No dominated strategies.

Example – Dominated Strategies

- Reduced payoff table:

	1	2	3
1	1	2	4
2	1	0	5

- Because both players are assumed to be rational, Player 2 also can deduce that Player 1 has only these 2 strategies remaining.
  - Player 2 now does have a dominated strategy.
  - Strategy 3 is dominated by both Strategy 1 and Strategy 2

Example – Dominated Strategies

- Reduced payoff table:

	1	2
1	1	2
2	1	0

  - Strategy 2 for Player 1 is dominated by Strategy 1.

Example – Dominated Strategies

- Reduced payoff table:

	1	2
1	1	2

- Strategy 2 for Player 2 is now dominated by Strategy 1.
- Both players should select their Strategy 1.
  - Player 1 will receive a payoff of 1 from Player 2.

Dominated Strategies

- Very useful for reducing the size of the payoff table.
- Can also help in identifying the optimal solution for the game.

Game Theory

- Value of the game.
  - Payoff to one player when both players play optimally.
  - A game with a value of zero is called fair game.

Example

- Suppose the payoff table for Player 1 is as follows.

Strategy		**Player 2**		
		1	2	3
Player 1	1	-3	-2	6
	2	2	0	2
	3	5	-2	-4

- No dominated strategies.

Example – Player 1 Thinking

- Select 1: could win 6, but could lose as much as 3.
  - Player 2 is rational and such will seek a strategy that will protect him from large payoffs to Player 1.
- Select 3: could win 5, but could lose as much as 4.
  - Player 2 is rational and such will seek a strategy that will protect him from large payoffs to Player 1.

Strategy		**Player 2**		
		1	2	3
Player 1	1	-3	-2	6
	2	2	0	2
	3	5	-2	-4

Example – Player 1 Thinking

- Select 2: guaranteed to not to lose anything, and even could win something.
  - Rational choice for Player 1 against his rational opponent.
  - Provides the best guarantee (a payoff of 0).

Strategy		**Player 2**		
		1	2	3
Player 1	1	-3	-2	6
	2	2	0	2
	3	5	-2	-4

Example – Player 2 Thinking

- Select 1: could lose as much as 5.
- Select 3: could lose as much as 6.
- Select 2: guaranteed to break even.
  - Rational choice for Player 2.
  - Seek the best guarantee against a rational opponent.

Strategy		**Player 2**		
		1	2	3
Player 1	1	-3	-2	6
	2	2	0	2
	3	5	-2	-4

Example – Minimax Criterion

- Minimize the maximum losses whenever the resulting choice of strategy cannot be exploited by his opponent to improve their position.
  - Select a strategy that would be best even if the selection were being announced to the opponent before the opponent chooses a strategy.

Strategy		**Player 2**		
		1	2	3
Player 1	1	-3	-2	6
	2	2	0	2
	3	5	-2	-4

Example – Minimax Criterion

- Player 1 should select the strategy whose minimum payoff is largest.
  - Maximin strategy.
- Player 2 should select the strategy whose maximum payoff to Player 1 is the smallest.
  - Minimax strategy.

Strategy		Player 2			
		1	2	3	Min
Player 1	1	-3	-2	6	-3
	2	2	0	2	0 Maximin
	3	5	-2	-4	-4
	Max	5	0 Minimax	6	

Example – Saddle Point

- Same entry in the payoff table yields both the maximin and the minimax values.
  - Both the minimum in its row and maximum in its column.
- Neither player can take advantage of the opponent’s strategy to improve own position.
- Neither player has any motive to consider changing strategies, either to take advantage of the opponent or to prevent the opponent to take advantage.
- Stable position (also called equilibrium position) in which players should use their maximin and minimax strategies, respectively.

Example

- Suppose the payoff table for Player 1 changed to:

Strategy		Player 2		
		1	2	3
Player 1	1	0	-2	2
	2	5	4	-3
	3	2	3	-4

- No dominated strategies.
- Players attempt to apply minimax criterion.

Example – Apply Minimax Criterion

- Player 1 can guarantee a loss of no more than 2 by playing Strategy 1.
- Player 2 can guarantee a loss of no more than 2 by playing Strategy 3.
  - No saddle point.

Strategy		Player 2			
		1	2	3	Min
Player 1	1	0	-2	2	-2 Maximin
	2	5	4	-3	-3
	3	2	3	-4	-4
	Max	5	4	2 Minimax	

Example – Apply Minimax Criterion

- If both players play their minimax strategies.
  - Player 1 would win 2 from Player 2, which would make Player 2 unhappy.
  - Because Player 2 is rational and can therefore foresee the outcome, he would then conclude that he can do much better, actually winning 2 rather than losing 2, by choosing Strategy 2 instead.
  - Because Player 1 is also rational, he would anticipate the switch and conclude that he can improve considerably, from -2 to 4, by changing to Strategy 2.
  - Realizing this, Player 2 would then consider switching back to Strategy 3 to convert a loss of 4 to a gain of 3.
  - The possibility of a switch would cause Player 1 to consider again using Strategy 1, after which the whole cycle would start over again.

Example – Unstable Solution

- Even though this game is played only once, any tentative choice of a strategy leaves the player with a motive to consider changing strategies, either to take advantage of his opponent or to prevent the opponent from taking advantage of him.
- The suggested solution (Player 1 plays Strategy 1 and Player 2 plays Strategy 3) is an unstable solution.
  - Need a more satisfactory solution.

Example – Finding a Solution

- Whenever one player's strategy is predictable, the opponent can take advantage of this information to improve his position.
  - Therefore, an essential feature of a rational plan for playing a game such as this one is that neither player should be able to deduce which strategy the other one will use.
  - Rather than applying some known criterion for determining a single strategy that will definitely be used, it is necessary to choose among alternative acceptable strategies on some kind of random basis.
  - Neither player knows in advance which of his own strategies will be used, let alone what his opponent will do.

Games With Mixed Strategies

- Whenever a game does not have a saddle point, game theory advises each player to assign a probability distribution over the set of strategies.

x_i = probability that player 1 will use strategy i (i = 1, 2, …, m)

m = number of available strategies to player 1

$x_1 + x_2 + \ldots + x_m = 1$; $x_1, x_2, \ldots x_m \geq 0$

y_i = probability that player 2 will use strategy j (j = 1, 2, …, n)

n = number of available strategies to player 2

$y_1 + y_2 + \ldots + y_m = 1$; $y_1, y_2, \ldots y_m \geq 0$

($x_1, x_2, \ldots x_m$) and ($y_1, y_2, \ldots y_m$) are called mixed strategies

Also, original strategies are called pure strategies

Games With Mixed Strategies

- When the game is actually played, each player uses one of the pure strategies.
- But, the pure strategy is chosen by using some random device to obtain a random observation from the probability distribution specified by the mixed strategy, where the observation indicates which particular pure strategy to use.

Games With Mixed Strategies

- Mixed strategies:
  - $(x_1, x_2, \dots x_m) = (0.5, 0.5, 0)$
  - Player 1 randomly selects between strategies 1 and 2.
  - $(y_1, y_2, \dots y_m) = (0, 0.5, 0.5)$
  - Player 2 randomly selects between strategies 2 and 3.

Strategy		**Player 2**		
		1	2	3
Player 1	1	0	-2	2
	2	5	4	-3
	3	2	3	-4

Games With Mixed Strategies

- There is no completely satisfactory measure of performance available for evaluating mixed strategies.
  - An expected one is the expected value (payoff):
- Expected payoff for player 1 = $\sum_{i=1}^{m}\sum_{j=1}^{n} p_{ij} x_i y_j$
  - p_{ij} = payoff if player 1 uses pure strategy i and player 2 uses pure strategy j.
  - Expected payoff for player 1 = (0.25)(-2+2+4-3) = 0.25
  - Average payoff of the game is played many times.
  - No information about the risk involved.

Games With Mixed Strategies

- Extension of Minimax Criterion to games that lack a saddle point.
  - A given player should select the mixed strategy that minimizes the maximum expected loss to himself (from Player 2 perspective).
    - Maximum expected loss = largest possible expected payoff that can be administered by any of the opponent’s mixed strategies.
  - A given player should select the mixed strategy that maximizes the minimum expected payoff to the player (from Player 1 perspective).
    - Minimum expected payoff = smallest possible expected payoff that can result from any mixed strategy with which the opponent can counter.

More on Minimax Criterion

- Odds and Evens Game.
  - No saddle point – unstable solution.
  - Rational way to play is to make the choice completely randomly each time.

Strategy		Player 2		
		1	2	Min
Player 1	1	1	-1	-1
	2	-1	1	-1
	Max	1	1	

More on Minimax Criterion

- Example:
  - Value of the game = 3

Strategy		Player 2		
		1	2	Min
Player 1	1	3 Saddle point	5	3 Maximin
	2	1	-2	-2
	Max	3 Minimax	5	

More on Mixed Strategies

- If X chooses X_1, then P percent of the time the payoff for Y will be Y_1, and (1-P) percent of the time, the payoff will be Y_2
- If X chooses X_2, then P percent of the time the payoff for Y will be Y_1, and (1-P) percent of the time, the payoff will be Y_2
- If the expected values for Y are the same, then the expected value for Y will not depend on the strategy chosen by X.

Strategy		**Player *Y***	
		Y_1 P	Y_2 $1-P$
Player *X*	X_1 Q	4	2
	X_2 $1-Q$	1	10

More Complicated Games

- Two-person, constant sum-game.
  - Sum of payoffs to the two player is a fixed constant (positive or negative) regardless which combination of strategies is selected.
  - The two players may share some reward or some cost for participating in the game.
  - Zero-sum game approach applies.
- *n*-person game.
  - More than 2 players may participate in the game.
  - In many competitive situations, frequently more than 2 competitors are involved.
  - Theory is more complicated and less satisfactory.

More Complicated Games

- Nonzero-sum game.
  - Sum of payoffs to the players need not be 0 (or any other fixed constant).
  - Many competitive situations include noncompetitive aspects that contribute to the mutual advantage or mutual disadvantage of the players.
  - Size of mutual gain (or loss) to players depends on the combination of strategies chosen.

More Complicated Games

- Nonzero-sum game.
  - Because mutual gain is possible, nonzero-sum games are further classified in terms of the degree to which the players are permitted to cooperate.
  - At one extreme is the non-cooperative game, where there is no preplay communication between the players.
  - At the other extreme is the cooperative game, where preplay discussions and binding agreements are permitted.
  - Competitive situations involving trade regulations between countries, or collective bargaining between labor and management.
  - For two or more players, cooperative games allow some of the players to form coalitions.

More Complicated Games

- Infinite games.
  - The players have an infinite number of pure strategies available to them.
  - The strategy can be represented by a continuous decision variable.

Prisoner's Dilemma

Natural Defection

A game theory paradox called the Prisoner's Dilemma illustrates why the existence of cooperation in nature is unexpected. Two people face jail sentences for conspiring to commit a crime. Their sentences depend on whether they elect to cooperate and remain silent or defect and confess to the crime [*see payoff table below*]. Because neither knows what the other will do, the rational choice—the one that always offers the better payoff—is to defect.

		INDIVIDUAL 2	
		COOPERATE (remain silent)	**DEFECT** (confess)
INDIVIDUAL 1	**COOPERATE** (remain silent)	2 years in jail 2 years in jail	4 years in jail 1 year in jail
	DEFECT (confess)	1 year in jail 4 years in jail	3 years in jail 3 years in jail